

Climate 5000 Series - Gen 4

9K BTU Single Zone Max Performance System

Wall Mounted IDU

Engineering
 Submittal
 Sheet



BOSCH

Overview and Project Information



Project name	
Location	
Architect	
Engineer	
General contractor	
Reference #	

9K Btu Single Zone Max Performance System - Wall Mounted IDU

Indoor part number	8-733-962-672
Indoor model number	BMS500-AAU009-1AHWXD
Outdoor part number	8-733-962-688
Outdoor model number	BMS500-AAS009-1CSXHD

Efficiency Data	AHRI 210/240 2023
SEER2	27.40
EER2	15.40
HSPF2 Region IV	12.5
HSPF2 Region V	9.5
COP(47F)	3.76
COP (5F)	1.80

Performance Data		AHRI 210/240 2023
Cooling	Capacity Range	3800~13500
	Rated Capacity	9000
	Current	4.1
	Input Power	580
Heating	Capacity Range	3800~16000
	Rated Capacity (47F)	12000
	Rated Capacity (5F)	11500
	Current	4.1
	Input Power	935

Electrical Data	
Power supply (V / Ø / Hz)	208-230 / 1 / 60
Min. circuit ampacity indoor/outdoor (A)	3.0 / 15.0
Max. fuse indoor/outdoor (A)	15.0 / 15.0

Compressor Data	
Type:	Rotary
Refrigerant oil/oil charge (ml)	VG74 / 350
Input (W)	885
Rated current (RLA/A)	6.0

Operation Range		
ODU	Cooling (°F / °C)	-22~122 / -30~50
	Heating (°F / °C)	-22~75 / -30~24
IDU	Cooling (°F / °C)	60~90 / 16~32
	Heating (°F / °C)	32~86 / 0~30

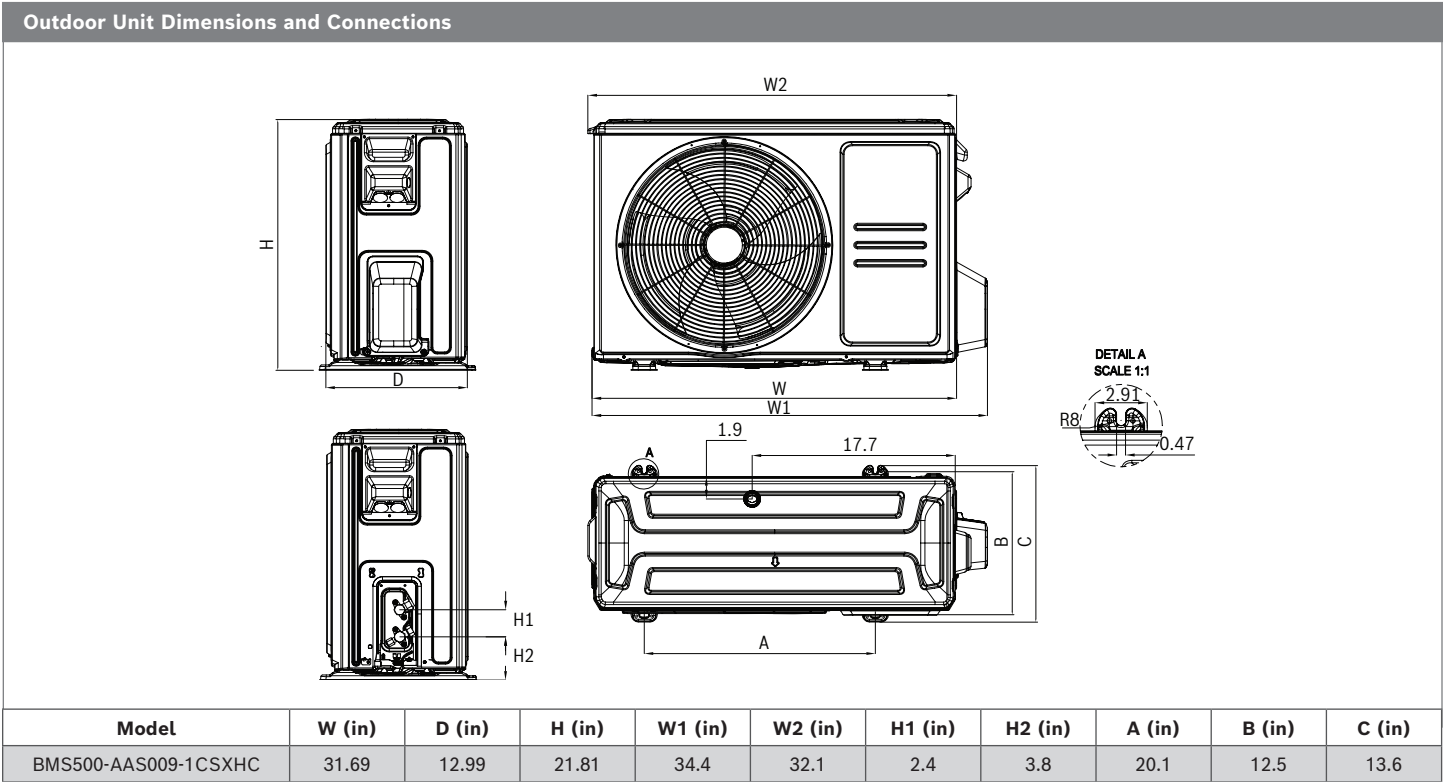
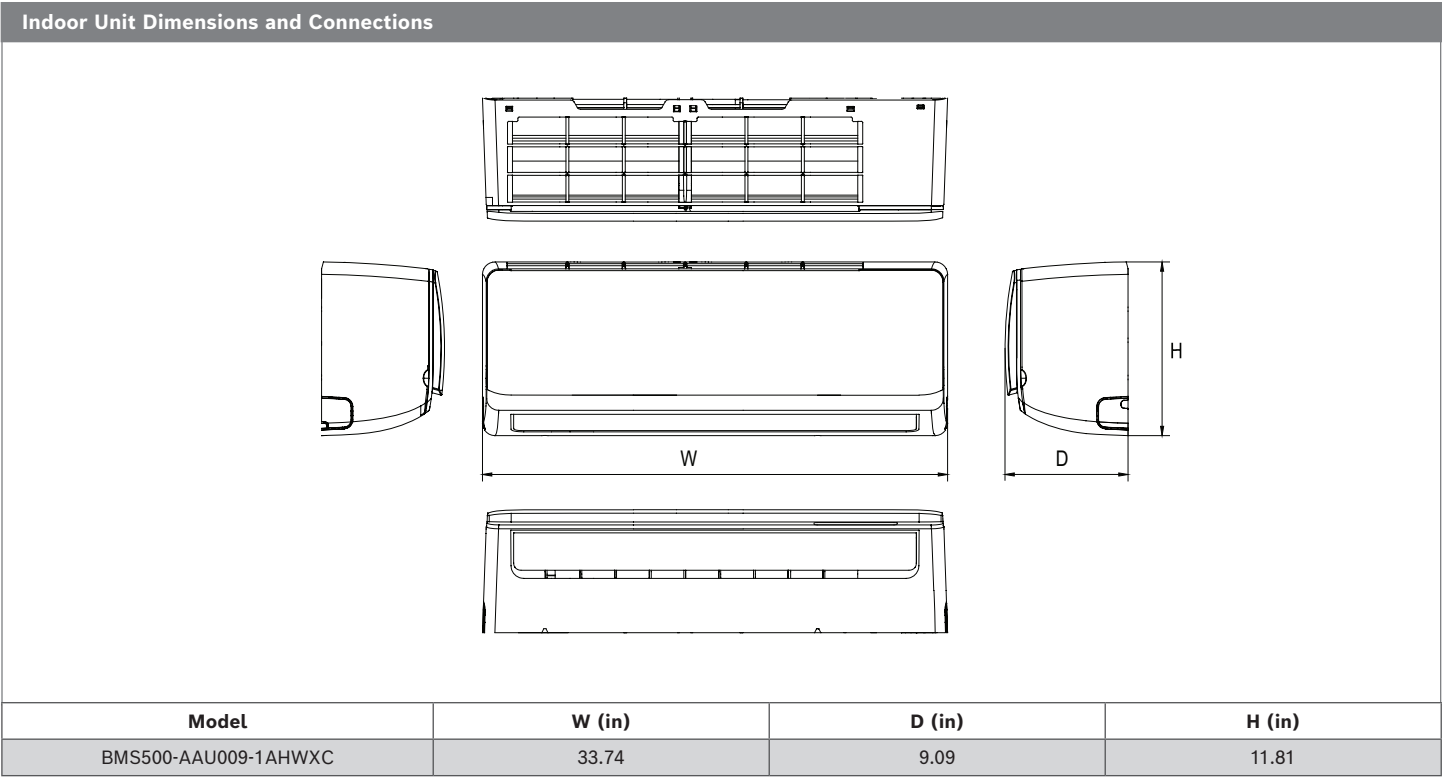
Piping	
Refrigerant precharge (oz/g)	35.27 / 1000
Length for precharge (ft/m)	24.6 / 7.5
Liquid side (in/mm)	1/4" / Ø6.35
Gas side (in/mm)	3/8" / Ø9.52
Additional Charge of Refrigerant (oz/ft)/(g/m)	0.16 / 15
Max. refrigerant pipe length (ft/m)	82 / 25
Max. difference in height (ft/m)	49.2 / 15

System Data	
Refrigerant type	
R454B	
Design pressure (PSIG)	
550 / 340	
Dehumidification (pts/hr)	
0.63	
IDU	Fan motor model
	ZKFP-20-8-6-7
	Fan motor qty (ea)
	1
	Fan motor input (W)
	50.0
	Fan speed (RPM)
	1100 / 850 / 700
	Air flow, Turbo / H / M / L / Si (CFM)
441.4 / 323.7 / 229.6 / 176.6 / 147.2	
Approx. Air Throw Range ft (m)	
23.0 / 7	
Noise level, Turbo / H / M / L / Si (dB(A))	
NA / 38 / 33 / 20.5 / 19.5	
Net weight (lbs / kg)	
23.37 / 10.6	
Gross weight (lbs / kg)	
29.98 / 13.6	
ODU	Fan motor model
	ZKFN-34-10-1-3
	Fan motor qty (ea)
	1
	Fan motor input (W)
	50
	Fan motor speed, H / / L (RPM)
	850 / 680 / 450
	Air flow, Max (CFM)
1470.6	
Noise level, Max (dB(A))	
56	
Net weight (lbs / kg)	
72.75 / 33	
Gross weight (lbs / kg)	
78.7 / 35.7	

Accessories	
BMS-WT2-XXD (8733962707)	Wired Wall Thermostat
BMS-DRY-XXD (8733969263)	24V Interface
BMS-PK1-7X1 (8733951010)	25ft Lineset

Standard Parts	
8-733-969-088	Remote Control

Climate 5000 Series - Gen 4
9K Btu Single Zone Max Performance System - Wall Mounted IDU



Climate 5000 Series - Gen 4

9K Btu Single Zone Max Performance System - Wall Mounted IDU


BOSCH

Cooling Performance - Indoor Dry Bulb Temperature at 441 CFM																					
ODU DB(F)	IDU WB (F)	60.8°F(16.0°C)					64.4°F(18.0°C)					67°F(19.4°C)					71.6°F(22.0°C)				
	IDU DB (F)	73.4°F (23.0°C)	77.0°F (25.0°C)	80.0°F (26.7°C)	80.6°F (27.0°C)	84.2°F (29.0°C)	73.4°F (23.0°C)	77.0°F (25.0°C)	80.0°F (26.7°C)	80.6°F (27.0°C)	84.2°F (29.0°C)	73.4°F (23.0°C)	77.0°F (25.0°C)	80.0°F (26.7°C)	80.6°F (27.0°C)	84.2°F (29.0°C)	73.4°F (23.0°C)	77.0°F (25.0°C)	80.0°F (26.7°C)	80.6°F (27.0°C)	84.2°F (29.0°C)
-22°F (-30.0°C)	TC	9076	9071	9201	9239	9493	9469	9461	9455	9454	9447	9752	9740	9731	9729	9720	10287	10284	10263	10259	10237
	S/T	0.80	0.91	0.99	0.99	0.99	0.64	0.75	0.84	0.86	0.97	0.55	0.65	0.74	0.75	0.86	0.39	0.47	0.55	0.57	0.67
	PI	0.36	0.36	0.36	0.36	0.36	0.36	0.36	0.36	0.36	0.36	0.36	0.36	0.36	0.36	0.36	0.36	0.36	0.36	0.36	0.36
-13°F (-25.0°C)	TC	9832	9827	9968	10009	10284	10258	10250	10243	10242	10234	10565	10552	10542	10540	10530	11144	11141	11118	11114	11090
	S/T	0.80	0.91	1.00	1.00	1.00	0.65	0.75	0.85	0.86	0.97	0.55	0.65	0.74	0.76	0.86	0.39	0.47	0.56	0.57	0.67
	PI	0.37	0.37	0.37	0.37	0.37	0.37	0.37	0.37	0.37	0.37	0.37	0.37	0.37	0.37	0.37	0.37	0.37	0.37	0.37	0.37
-4°F (-20.0°C)	TC	10588	10583	10734	10779	11075	11047	11038	11031	11030	11021	11377	11363	11352	11351	11340	12001	11998	11973	11969	11943
	S/T	0.80	0.92	1.00	1.00	1.00	0.65	0.76	0.85	0.87	0.98	0.55	0.65	0.74	0.76	0.86	0.39	0.48	0.56	0.57	0.67
	PI	0.39	0.39	0.39	0.39	0.39	0.38	0.38	0.39	0.39	0.39	0.38	0.38	0.38	0.38	0.38	0.39	0.39	0.39	0.39	0.39
0°F (-17.8°C)	TC	10940	10933	11136	11182	11490	11409	11396	11387	11386	11500	11744	11727	11713	11711	11705	12376	12373	12372	12373	12381
	S/T	0.81	0.93	1.00	1.00	1.00	0.66	0.77	0.86	0.88	1.00	0.55	0.66	0.75	0.77	0.87	0.39	0.48	0.57	0.58	0.68
	PI	0.39	0.39	0.39	0.39	0.39	0.39	0.39	0.39	0.39	0.39	0.39	0.39	0.39	0.39	0.39	0.39	0.39	0.39	0.39	0.39
5°F (-15.0°C)	TC	11253	11292	11550	11603	11961	11781	11784	11785	11797	11973	12158	12161	12163	12164	12180	12871	12867	12874	12875	12881
	S/T	0.82	0.94	1.00	1.00	1.00	0.66	0.78	0.87	0.88	1.00	0.56	0.67	0.76	0.78	0.88	0.39	0.48	0.57	0.59	0.69
	PI	0.39	0.39	0.39	0.39	0.40	0.39	0.40	0.40	0.40	0.40	0.40	0.40	0.40	0.40	0.40	0.41	0.41	0.41	0.41	0.41
14°F (-10.0°C)	TC	11391	11385	11617	11672	12034	11956	11945	11937	11936	12048	12366	12350	12338	12336	12323	13149	13145	13119	13115	13087
	S/T	0.81	0.93	1.00	1.00	1.00	0.65	0.76	0.86	0.87	1.00	0.55	0.66	0.75	0.76	0.87	0.39	0.48	0.56	0.57	0.67
	PI	0.41	0.41	0.41	0.41	0.41	0.41	0.41	0.41	0.41	0.41	0.41	0.41	0.41	0.41	0.41	0.41	0.41	0.41	0.41	0.41
17°F (-8.3°C)	TC	11409	11447	11711	11768	12151	11977	11982	11984	11995	12164	12386	12391	12396	12396	12412	13173	13169	13167	13170	13181
	S/T	0.82	0.93	1.00	1.00	1.00	0.66	0.77	0.87	0.88	1.00	0.55	0.66	0.76	0.77	0.87	0.39	0.48	0.57	0.58	0.68
	PI	0.42	0.42	0.42	0.42	0.42	0.42	0.42	0.42	0.42	0.42	0.42	0.42	0.42	0.42	0.42	0.42	0.42	0.42	0.42	0.42
23°F (-5.0°C)	TC	11687	11737	12041	12100	12499	12279	12282	12284	12300	12512	12702	12707	12710	12710	12731	13506	13502	13512	13514	13521
	S/T	0.83	0.95	1.00	1.00	1.00	0.66	0.78	0.88	0.89	1.00	0.56	0.67	0.76	0.78	0.89	0.39	0.48	0.57	0.59	0.69
	PI	0.46	0.46	0.46	0.46	0.47	0.46	0.46	0.46	0.46	0.47	0.47	0.47	0.47	0.47	0.47	0.48	0.48	0.48	0.48	0.48
35°F (1.7°C)	TC	12356	12356	12429	12464	12866	12934	12933	12934	12934	12969	13338	13338	13338	13339	13340	14090	14086	14085	14085	14086
	S/T	0.78	0.88	0.97	0.99	1.00	0.63	0.73	0.82	0.83	0.93	0.54	0.63	0.72	0.73	0.83	0.39	0.46	0.54	0.56	0.65
	PI	0.51	0.51	0.52	0.52	0.54	0.54	0.54	0.54	0.54	0.54	0.57	0.57	0.57	0.57	0.57	0.62	0.62	0.62	0.62	0.62
41°F (5.0°C)	TC	12244	12244	12342	12427	12795	12813	12813	12814	12814	12859	13213	13212	13213	13213	13214	13955	13951	13950	13950	13951
	S/T	0.78	0.89	0.98	1.00	1.00	0.64	0.74	0.83	0.84	0.94	0.54	0.64	0.72	0.74	0.84	0.39	0.47	0.55	0.56	0.66
	PI	0.50	0.50	0.51	0.51	0.53	0.53	0.53	0.53	0.53	0.54	0.56	0.56	0.56	0.56	0.56	0.61	0.61	0.61	0.61	0.61
47°F (8.3°C)	TC	12131	12139	12302	12357	12723	12694	12693	12694	12694	12752	13088	13087	13088	13088	13088	13821	13817	13816	13816	13817
	S/T	0.79	0.90	1.00	1.00	1.00	0.64	0.75	0.84	0.85	0.96	0.54	0.65	0.73	0.75	0.85	0.39	0.47	0.55	0.57	0.67
	PI	0.50	0.50	0.50	0.51	0.53	0.53	0.53	0.53	0.53	0.53	0.55	0.55	0.55	0.55	0.55	0.60	0.60	0.60	0.60	0.60
50°F (10.0°C)	TC	11878	12205	12617	12690	13182	12549	12556	12643	12682	13195	13038	13055	13062	13063	13189	13937	13974	14010	14014	14032
	S/T	0.90	1.00	1.00	1.00	1.00	0.71	0.85	0.96	0.99	1.00	0.59	0.72	0.83	0.85	0.99	0.40	0.50	0.60	0.62	0.74
	PI	0.56	0.56	0.56	0.56	0.56	0.56	0.56	0.56	0.56	0.56	0.56	0.56	0.56	0.56	0.56	0.56	0.56	0.56	0.56	0.56
59°F (15.0°C)	TC	11771	12184	12616	12693	13210	12454	12471	12613	12707	13223	12968	12981	12986	13005	13234	13925	13953	13981	13984	13995
	S/T	0.92	1.00	1.00	1.00	1.00	0.73	0.86	1.00	1.00	1.00	0.60	0.73	0.85	0.86	1.00	0.40	0.51	0.61	0.63	0.76
	PI	0.56	0.57	0.57	0.57	0.57	0.57	0.57	0.57	0.57	0.57	0.57	0.57	0.57	0.57	0.57	0.57	0.57	0.57	0.57	0.57
68°F (20.0°C)	TC	11659	12156	12607	12687	13226	12343	12370	12621	12701	13239	12876	12886	12909	12920	13250	13874	13900	13922	13924	13931
	S/T	0.95	1.00	1.00	1.00	1.00	0.74	0.89	1.00	1.00	1.00	0.61	0.75	0.87	0.89	1.00	0.40	0.51	0.63	0.65	0.78
	PI	0.57	0.57	0.58	0.58	0.58	0.58	0.58	0.58	0.58	0.58	0.58	0.58	0.58	0.58	0.58	0.59	0.59	0.59	0.59	0.59
77°F (25.0°C)	TC	11538	12093	12566	12650	13215	12231	12270	12580	12664	13229	12785	12800	12832	12847	13240	13809	13850	13883	13886	13899
	S/T	0.97	1.00	1.00	1.00	1.00	0.75	0.90	1.00	1.00	1.00	0.61	0.76	0.88	0.90	1.00	0.40	0.52	0.63	0.65	0.79
	PI	0.58	0.59	0.59	0.59	0.59	0.59	0.59	0.59	0.59	0.59	0.59	0.59	0.59	0.59	0.59	0.60	0.60	0.60	0.60	0.60
82°F (27.8°C)	TC	11530	12119	12608	12693	13274	12218	12266	12619	12707	13288	12787	12803	12842.76	12859	13300	13825	13877	13913	13917	13932
	S/T	0.99	1.00	1.00	1.00	1.00	0.76	0.91	1.00	1.00	1.00	0.62	0.77	0.89	0.92	1.00	0.40	0.52	0.64	0.66	0.80
	PI	0.66	0.66	0.67	0.67	0.67	0.66	0.66	0.66	0.66	0.67	0.67	0.67	0.67	0.67	0.67	0.67	0.67	0.67	0.67	0.67
86°F (30.0°C)	TC	11639	12220	12720	12809	13405	12275	12331	12734	12823	13419	12856	12871	12918	12937	13431	13918	13972	14005	14009	14022
	S/T	1.00	1.00	1.00	1.00	1.00	0.77	0.93	1.00	1.00	1.00	0.62	0.78	0.91	0.93	1.00	0.40	0.52	0.64	0.66	0.81
	PI	0.80	0.81	0.81	0.81	0.82	0.81	0.81	0.81	0.81	0.82	0.81	0.81	0.81	0.81	0.82	0.82	0.82	0.82	0.82	0.82
95°F (35.0°C)	TC	11798	11862	12245	12318	12805	12539	12546													

Climate 5000 Series - Gen 4

9K Btu Single Zone Max Performance System - Wall Mounted IDU



Heating Performance - Indoor Dry Bulb Temperature at 441 CFM										
Outdoor DB(°F)	TC Total Heating Capacity(BTU/hr)					PI Power Input(kW)				
	Indoor Conditions(DB °F)					Indoor Conditions(DB °F)				
	60.8°F (16.0°C)	68°F (20.0°C)	70°F (21.1°C)	71.6°F (22.0°C)	75.2°F (24.0°C)	60.8°F (16.0°C)	68°F (20.0°C)	70°F (21.1°C)	71.6°F (22.0°C)	75.2°F (24.0°C)
-22°F(-30.0°C)	5238	5133	5107	5087	5041	1.34	1.35	1.35	1.35	1.36
-13°F(-25.0°C)	6157	6034	6004	5980	5926	1.43	1.44	1.44	1.45	1.45
-4°F(-20.0°C)	8287	8121	8080	8048	7976	1.57	1.59	1.59	1.59	1.60
0°F(-17.8°C)	9920	9721	9673	9634	9547	1.63	1.64	1.64	1.65	1.65
5°F(-15.0°C)	11961	11722	11663	11617	11512	1.69	1.71	1.71	1.71	1.72
14°F(-10.0°C)	12292	12046	11986	11938	11831	1.62	1.64	1.64	1.64	1.65
17°F(-8.3°C)	12402	12154	12093	12045	11937	1.60	1.62	1.62	1.62	1.63
22°F(-5.6°C)	12996	12737	12673	12622	12509	1.60	1.61	1.61	1.62	1.62
27°F(-2.8°C)	13591	13319	13252	13199	13080	1.59	1.60	1.61	1.61	1.62
32°F(0.0°C)	14185	13901	13832	13776	13652	1.59	1.60	1.60	1.60	1.61
35°F(1.7°C)	14541	14251	14179	14123	13995	1.58	1.60	1.60	1.60	1.61
37°F(2.8°C)	14878	14581	14508	14450	14320	1.54	1.56	1.56	1.56	1.57
42°F(5.6°C)	15720	15406	15328	15267	15130	1.45	1.46	1.46	1.46	1.47
44.6°F(7.0°C)	16057	15736	15657	15594	15454	1.41	1.42	1.42	1.42	1.43
47°F(8.3°C)	16562	16231	16149	16085	15940	1.35	1.36	1.36	1.37	1.37
52°F(11.1°C)	15865	15548	15470	15408	15269	1.22	1.23	1.23	1.23	1.24
57°F(13.9°C)	15168	14865	14791	14731	14599	1.08	1.09	1.09	1.10	1.10
59°F(15.0°C)	14890	14592	14519	14461	14331	1.03	1.04	1.04	1.04	1.05
62°F(16.7°C)	14472	14182	14111	14055	13928	0.95	0.96	0.96	0.96	0.97
64.4°F(18.0°C)	14137	13854	13785	13730	13606	0.89	0.89	0.90	0.90	0.90
69.8°F(21.0°C)	13385	13117	13051	12999	12882	0.74	0.75	0.75	0.75	0.76
75.2°F(24.0°C)	12632	12380	12318	12268	12158	0.60	0.60	0.61	0.61	0.61

Bolded values were tested. All other values are simulated and for reference only.